

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the operation of stylus Probe instruments for measuring surface roughness.

Q.24 Discuss the concept and working principles of Tool Maker's Microscope.

Q.25 Write short notes on:

- Surface Texture of Measurement
- Co-ordinate Measuring Machine

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2nd Year / Advance Diploma in Tool and Die Making
Subject : Engineering Metrology

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which relationship generally holds for sensitivity and range of measuring instruments?

- a) direct b) normal
c) linear d) inverse

Q.2 Which errors are inevitable in instruments?

- a) systematic b) calibration
c) ransom d) parasitic

Q.3 A Set of 16 angle gauges can form angles between 0 degree and 90 degree in intervals of

- a) 1" b) 5"
- c) 1' d) 1 degree

Q.4 Which is related with measurements using light waves

- a) lapped surface b) optical flats
- c) comparators d) spirit level

Q.5 In a limit system, a series of tolerances are arranged to suit

- a) a specific range of size
- b) cost
- c) desired function
- d) allowances, desired

Q.6 In hole basis system, basic size is equal to the

- a) maximum hole size b) minimum hole size
- c) average hole size d) none

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define sensitivity.

Q.8 What is feeler gauge?

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Q.9 Define Measurements.

Q.10 Define Fits.

Q.11 Define GO GAUGES.

Q.12 Define micrometer.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Write advantages and disadvantages of inspection.

Q.14 Discuss the concept of Measuring instruments.

Q.15 Write the precautions to be taken in measurements.

Q.16 Explain the Selection of limits.

Q.17 Explain the Snap Gauges.

Q.18 Discuss the Ring Gauges.

Q.19 Explain Bore Gauges.

Q.20 Discuss the surface Irregularities.

Q.21 Explain with diagram 'V-BLOCK'.

Q.22 Discuss the concept of Interchangeability.

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